**10** Ultrasound and X-rays are both types of wave that are used in medical diagnosis to form images of internal body structures.

**(a)** Complete Table 10.1 to state, for each type of wave:

- the method of production of the wave
- whether the wave that is detected and used to form the image is the wave that has been absorbed, reflected or transmitted by the internal body structure.

**Table 10.1**

| | ultrasound | X-rays |
|---|---|---|
| method of production | ..... ..... | ..... ..... |
| detected wave (absorbed, reflected or transmitted) | ..... | ..... |

[4]

**(b) (i)** For one type of wave passing through tissue, the wave has 72% of its initial intensity after it has passed through 6.2 cm of the tissue.

Calculate the linear attenuation coefficient $\mu$ of the tissue for this wave.

$\mu$ = ..... $cm^{-1}$ [2]

**(ii)** Another wave of the same type as in **(b)(i)** passes through 9.3 cm of the same tissue.

Calculate the percentage of the initial intensity of the wave that is attenuated by the tissue.

percentage attenuated = ..... % [2]

[Total: 8]

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